Cardiometabolic Breakthroughs: A Thematic Investment Strategy

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Executive Summary

This project presents a thematic equity investment strategy focused on cardiometabolic innovation, developed to bridge my background in biophysics with quantitative portfolio construction and investment analysis.

The investment rationale is based on structural trends including ageing populations, increasing prevalence of obesity and diabetes, and rising healthcare costs. These conditions are chronic and interrelated, creating sustained demand for therapeutic innovation. Recent breakthroughs, particularly in incretin-based treatments, have reshaped the treatment landscape, while the broader ecosystem spans pharmaceuticals, biotechnology, medical devices, life-science tools and manufacturing.

A universe of 50 publicly listed companies was constructed using healthcare funds, thematic indices and bottom-up research. Each company was assessed through a proprietary thematic scoring framework incorporating exposure to cardiometabolic innovation, pipeline maturity, differentiation, validation, partnerships, revenue scale and growth. A portfolio of 16 equities was then selected while preserving the industry structure of the broader universe.

Portfolio weights were optimized using a constrained Bayesian maximum-entropy framework. The objective was to improve expected return and thematic exposure while minimizing unnecessary changes in allocation. Constraints were applied to position size, industry exposure, annualized volatility and Value-at-Risk to ensure diversification and risk control.

The portfolio delivered annualized returns of 24.3% over the period analyzed, compared with 4.9% for MSCI World, 4.2% for MSCI ACWI and 6.5% for the S&P 500. This outperformance was achieved with higher but controlled volatility and limited turnover, supporting the strategy's practical implementability. Concentration and exposure metrics confirm a diversified allocation consistent with the imposed constraints.

Factor analysis suggests that performance was not solely driven by general market exposure but also reflects a differentiated cardiometabolic innovation component. However, results should be interpreted with caution, as the period analyzed benefited from favorable sector-specific dynamics and does not fully capture the risks associated with clinical, regulatory and reimbursement uncertainty.

Overall, this project demonstrates the integration of scientific expertise, quantitative modelling and investment reasoning in a structured thematic approach. It is intended as an educational case study and not as investment advice.

Investment Case Snapshot

Theme
Exposure to cardiometabolic innovation driven by rising obesity, diabetes and cardiovascular disease, supported by ongoing therapeutic and technological progress.

Investment Rational
Structural, non-cyclical demand combined with rapid innovation in treatment and a broad ecosystem spanning pharmaceuticals, biotechnology, devices and manufacturing.

Approach
- 50-company investment universe built from funds, indices and sector research
- Proprietary thematic scoring framework (innovation, pipeline, validation, growth)
- Selection of 16 equities preserving industry structure
- Bayesian maximum-entropy portfolio optimization
- Explicit constraints on diversification and risk

Portfolio Construction
- Position size: 4% – 8%
- Industry exposure capped
- Volatility: < 20%
- 1-month 97.5% VaR: < 6%
- Low turnover due to entropy-based optimization

Key Results
- Annualized return: 24.3%
- Sharpe ratio: 1.3
- Outperformance vs MSCI World, MSCI ACWI and S&P 500
- Limited impact of transaction costs

Portfolio Characteristics
- Diversified across subsectors and geographies
- Stable exposure over time
- Balanced allocation between core and supporting positions
- Low concentration (HHI and concentration ratios)

Risk Profile
- Higher volatility typical of thematic equities
- Controlled tail risk
- Exposure to event-driven dynamics (clinical, regulatory, reimbursement)

Return Drivers
- Innovation cycles in cardiometabolic therapies
- Adoption and reimbursement expansion
- Growth in enabling and manufacturing infrastructure

Key Risks
- Clinical failure and scientific uncertainty
- Regulatory and pricing pressure
- Dependence on recent sector tailwinds

Conclusion
A structured thematic approach capturing cardiometabolic innovation can generate excess returns while maintaining diversification and risk discipline. Results remain sensitive to sector dynamics and should be interpreted cautiously.

Table of content

Executive Summary 1
Investment Case Snapshot 2
Table of content 3
Introduction 4
Thematic Rational 4
Scientific & Medical Background 4
Market Landscape 5
Investment Thesis 5
Thematic Fund Construction 6
  Thematic Universe 6
  Thematic Scoring 7
  Equity choice 7
  Portfolio optimization 7
  Method 7
    Constraints 8
    Benchmarks 8
Key performance 8
Performance Drivers 11
ESG & Impact Considerations 12
Conclusion 12

Introduction

This document summarizes a thematic equity research and portfolio-construction project focused on cardiometabolic innovation. The objective was to apply my background in biophysics, quantitative modelling and Python development to a finance-oriented investment problem.

The project was designed as a practical bridge between scientific research and investment management. It combines bottom-up thematic research, structured scoring, portfolio optimization, benchmark comparison and risk analysis. The emphasis is not on presenting a trade recommendation, but on demonstrating an investment process: defining a theme, building an investable universe, assessing company exposure, constructing a portfolio and evaluating performance within explicit risk constraints.

The core theme is cardiometabolic innovation. This includes pharmaceuticals, biotechnology, medical devices, life-science tools and manufacturing enablers linked to obesity, diabetes, cardiovascular disease and adjacent metabolic disorders. The theme was deliberately defined more broadly than GLP-1 obesity drugs alone, to capture a wider innovation ecosystem and reduce reliance on a single therapeutic class.

Full Python implementation is available on GitHub:
https://github.com/ValVal-Data/Thematic

Thematic Rationale

Cardiometabolic disease represents one of the most significant and persistent global health challenges. Rising life expectancy, urbanization and increasingly sedentary lifestyles have led to a structural increase in obesity, diabetes and cardiovascular disease. These conditions are chronic, often interrelated and associated with substantial long-term healthcare costs.

From an investment perspective, this creates a sustained and non-cyclical demand for more effective treatment, prevention and monitoring solutions. Healthcare systems have a strong economic incentive to address these diseases earlier, as late-stage complications are significantly more costly.

Recent therapeutic progress, particularly in incretin-based drugs, has reshaped expectations for the treatment of obesity and related metabolic disorders. However, the investment opportunity is broader than a single drug class. Innovation spans a wider ecosystem, including pharmaceutical companies, biotechnology firms, medical-device manufacturers, life-science tools providers and contract manufacturing organizations.

The thematic approach therefore seeks to capture:

- Long-term structural demand driven by demographics and lifestyle trends.
- Ongoing scientific innovation across multiple treatment modalities.
- Value creation across the full healthcare value chain.

At the same time, the theme presents material risks, including clinical failure, regulatory uncertainty, pricing pressure and reimbursement dynamics. These considerations justify a diversified approach across subsectors and company types.

Scientific & Medical Background

Cardiometabolic disease encompasses both cardiovascular and metabolic disorders. Cardiovascular conditions include diseases of the heart and blood vessels, while metabolic disorders affect processes such as glucose regulation, lipid metabolism and energy balance.

These conditions are typically chronic, progressive and often linked to each other. For example, obesity significantly increases the risk of diabetes and cardiovascular disease. This interdependence creates a broad and interconnected treatment landscape.

From a therapeutic perspective, innovation spans multiple approaches:
- Pharmaceutical treatments targeting metabolic pathways.
- Medical devices supporting monitoring, intervention or surgery.
- Enabling technologies used in drug development and manufacturing.

A key characteristic of this space is the long development cycle and high uncertainty associated with clinical trials. At the same time, successful innovation can lead to substantial commercial opportunities due to large patient populations and long treatment durations.

For investment purposes, the scientific complexity of the theme reinforces the importance of diversification across companies, technologies and stages of development.

Market Landscape

The cardiometabolic innovation landscape is characterized by a combination of large, established pharmaceutical companies and smaller, more specialized biotechnology firms, complemented by medical-device manufacturers and life-science tools providers.

Large pharmaceutical companies benefit from scale, global distribution capabilities and the financial resources required to conduct late-stage clinical trials and commercialization. They also play a central role in consolidating innovation through acquisitions of smaller biotechnology firms.

Biotechnology companies are often at the forefront of scientific innovation, particularly in early-stage drug discovery and novel therapeutic mechanisms. However, they are typically more volatile due to their reliance on clinical-trial outcomes and financing conditions.

Medical-device and life-science tools companies form an important part of the ecosystem by enabling diagnosis, monitoring, manufacturing and research. These segments tend to offer more stable revenue profiles, providing diversification within the theme.

The market structure therefore reflects a dynamic innovation cycle:
- Early-stage innovation driven by biotech.
- Scaling and commercialization led by large pharmaceutical firms.
- Supporting infrastructure provided by devices, tools and manufacturing.

From an investment perspective, this structure supports a diversified thematic approach across subsectors, balancing growth potential with risk control.

Investment Thesis

Cardiometabolic innovation represents a long-term structural opportunity within healthcare. Ageing populations, sedentary lifestyles, rising obesity prevalence and increasing diabetes incidence create a persistent medical and economic burden for healthcare systems. These diseases are often chronic, associated with comorbidities and costly to manage over long periods.

Recent therapeutic progress, particularly in incretin-based treatments, has materially changed the clinical and commercial outlook for obesity and related metabolic diseases. However, the investment case should not rely exclusively on one drug class. The broader cardiometabolic

ecosystem includes established pharmaceutical leaders, biotechnology innovators, medical-device companies, life-science tools providers and manufacturing specialists. Each contributes to a different part of the value chain. The thesis therefore combines three elements:

1. Structural demand: cardiometabolic diseases are large, chronic and under-treated globally.
2. Innovation momentum: new therapies and technologies are expanding treatment possibilities.
3. Value-chain breadth: opportunities extend beyond drug developers to manufacturing, diagnostics, devices and research tools.

The key investment risks are also material. These include clinical-trial failure, reimbursement uncertainty, regulatory intervention, pricing pressure, patent cycles and concentration risk around highly successful product categories. The portfolio construction process therefore incorporates explicit diversification and risk constraints.

Thematic Fund Construction

The construction of the thematic portfolio follows a structured process:

1. **Definition of the investment universe**
   A broad set of companies exposed to cardiometabolic innovation is identified using public fund holdings, thematic indices and sector research.
2. **Thematic scoring**
   Each company is evaluated using a structured scoring framework reflecting its exposure to the theme and the quality of its innovation.
3. **Equity selection**
   A subset of companies is selected based on their thematic score, while preserving the industry distribution of the broader universe.
4. **Portfolio optimization**
   Portfolio weights are determined through a constrained optimization framework balancing return, thematic exposure and risk control.
5. **Performance and risk evaluation**
   The resulting portfolio is analyzed in terms of returns, turnover, exposure stability, concentration and risk relative to benchmarks.

Thematic Universe

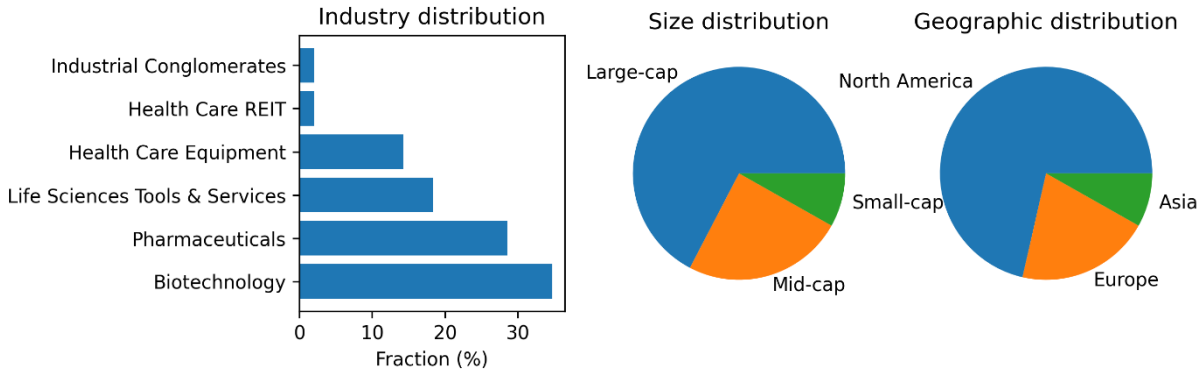


Figure 1 - Investment universe composition. Distribution of the 50-company universe by industry, market capitalisation and geography. The universe captures the breadth of the cardiometabolic innovation ecosystem while remaining concentrated in pharmaceutical and biotechnology sectors, with a strong representation of North America.

The investment universe consists of 50 publicly listed companies selected from healthcare funds, thematic indices and additional sector research. The universe includes pharmaceutical, biotechnology, medical-device and life-science tools companies directly involved in cardiometabolic innovation or supporting its development and manufacturing. Segments such

as digital health, nutrition and healthcare providers were excluded to maintain a clear and consistent exposure to therapeutic and technological innovation within cardiometabolic space. The resulting universe provides diversified exposure across regions, industries and company sizes.

Thematic Scoring

Companies are evaluated using a structured thematic scoring framework designed to capture both the degree of exposure to cardiometabolic innovation and the quality of that exposure. The scoring combines multiple dimensions, including:

- Thematic purity.
- Pipeline maturity and breadth.
- Differentiation and novelty of technology.
- Validation and partnerships.
- Revenue scale and growth.
- Role within the broader ecosystem.

Where precise quantitative data is not available, categorical assessments are used to ensure consistency and coverage across all companies. The objective is to balance exposure to highly innovative firms with more established players, while maintaining comparability across the universe.

Equity choice

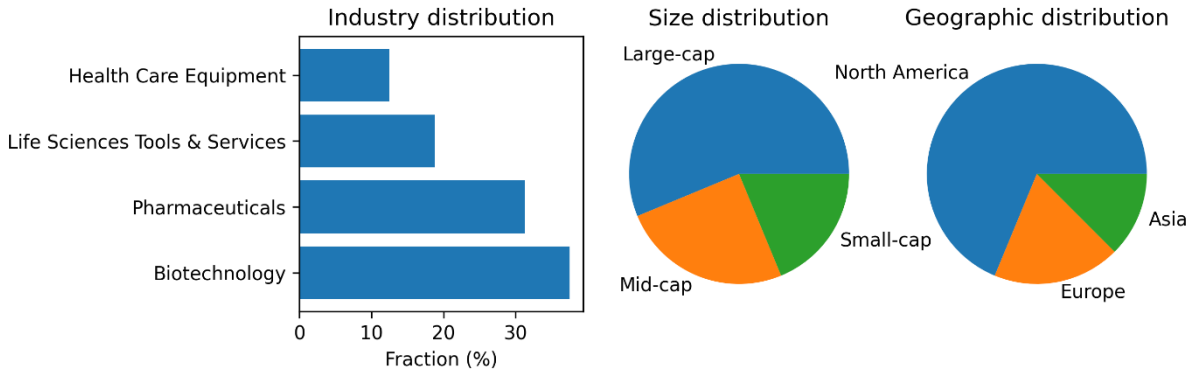


Figure 2 - Portfolio composition. Distribution of the selected 16-equity portfolio across industry, market capitalization and geography. The allocation preserves the industry structure of the broader universe while introducing a balanced exposure across company sizes and regions.

The final portfolio consists of 16 equities selected based on their thematic scores. Selection is performed within each industry group to preserve the overall industry structure of the investment universe. This approach ensures that the portfolio remains diversified and avoids excessive concentration in a single subsector, while still prioritizing exposure to the most relevant companies within the theme. The resulting portfolio includes a mix of large-cap and smaller companies, reflecting different stages of innovation and varying risk-return profiles

Portfolio optimization

Method

Portfolio weights are determined using a Bayesian maximum-entropy framework. The objective is to improve expected return and thematic exposure while minimizing unnecessary changes in portfolio weights.

The optimization starts from prior weights, defined as the most recent portfolio allocation after-market drift. The framework then identifies the smallest adjustment that improves the portfolio's characteristics, subject to defined constraints.

The objective function combines:
- A reward component based on expected return and thematic score.
- An entropy penalty measuring deviation from prior weights.

Return and thematic scores are normalized by their respective uncertainties, allowing the optimization to weigh both components consistently without relying on arbitrary parameters. The optimization is solved using Sequential Least Squares Programming, enabling the application of both equality and inequality constraints.

Constraints

Optimization is subject to the following constraints:
- Position size: 4% to 8%, to limit concentration risk
- Industry exposure: capped relative to the initial distribution, to maintain diversification
- Annualized volatility: below 20%, consistent with a thematic equity strategy
- One-month 97.5% Value-at-Risk: below 6%, to control tail risk

These constraints are designed to ensure that the portfolio remains diversified and aligned with an institutionally acceptable risk profile.

Benchmarks

Portfolio performance is evaluated against three benchmarks:

- MSCI World Index
- MSCI ACWI Index
- S&P 500 Index

MSCI World and MSCI ACWI provide broad global equity comparisons, while the S&P 500 offers a relevant reference given the portfolio's exposure to North American markets. These benchmarks help assess whether performance reflects general market movements or more theme-specific factors.

Key performance

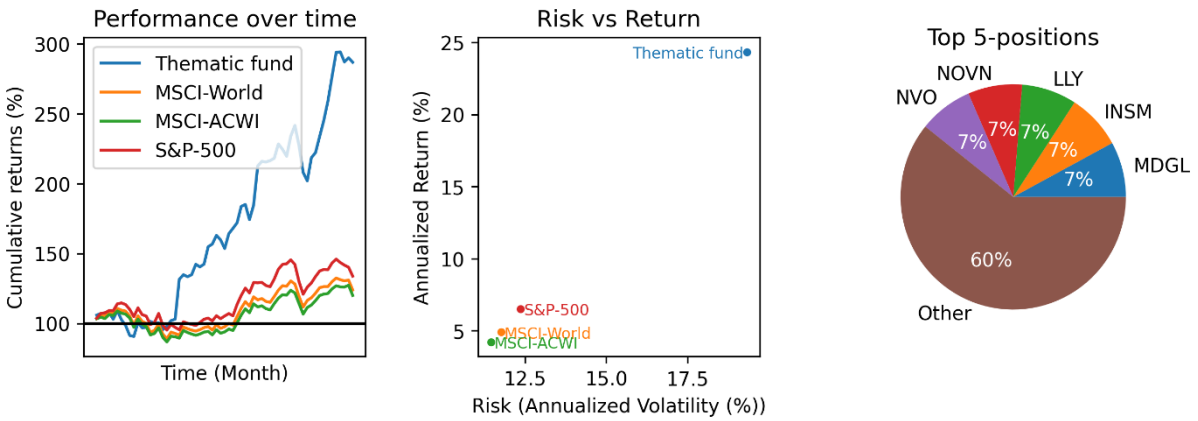


Figure 3 - Performance overview. Left: cumulative returns of the thematic portfolio versus global benchmarks. Center: risk-return profile showing higher returns accompanied by elevated but controlled volatility. Right: top positions illustrate a balanced allocation with no excessive concentration.

The portfolio demonstrates strong historical performance relative to global equity benchmarks over the analyzed period. Annualized returns reached 24.3%, compared with 4.9% for MSCI

World, 4.2% for MSCI ACWI and 6.5% for the S&P 500. This outperformance is accompanied by higher volatility, consistent with a thematic equity strategy focused on innovation-driven segments.

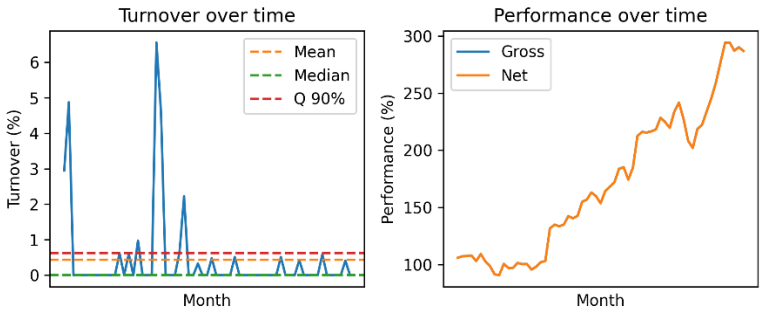


Figure 4 - Turnover and implementation profile. Left: portfolio turnover over time, reflecting the rebalancing process under the entropy-constrained optimization framework. Right: cumulative gross versus net returns, indicating that transaction costs have a limited impact on overall performance and supporting the implementability of the strategy.

Turnover remains limited over time, reflecting the structure of the optimization framework. The use of an entropy-based objective function penalizes unnecessary changes in portfolio weights, resulting in a stable allocation profile. Therefore, the difference between gross and net returns is relatively small, suggesting that transaction costs do not materially alter the overall performance profile. This supports the practical implementability of the strategy under the assumptions used.

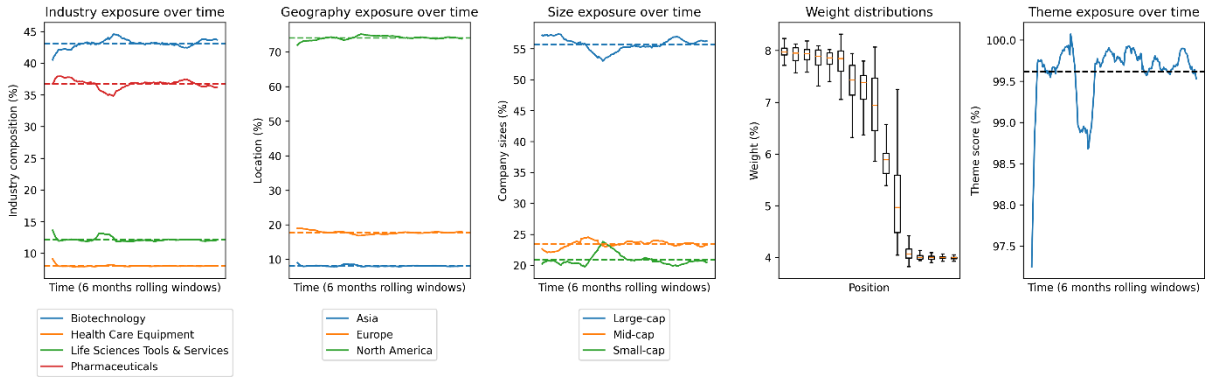


Figure 5 - Portfolio exposure dynamics. Industry, geographic and market-capitalization exposures over time. The portfolio maintains stable allocations consistent with imposed constraints, with weight distributions reflecting a structured balance between core holdings and supporting positions.

Exposure analysis indicates that the portfolio maintains a stable industry allocation, in line with the imposed constraints. Sector weights remain close to the initial universe distribution, while still allowing for allocation adjustments within the admissible range. Geographic and market-capitalization exposures also exhibit limited variation over time, despite not being explicitly constrained, highlighting the stabilizing effect of the overall optimization framework.

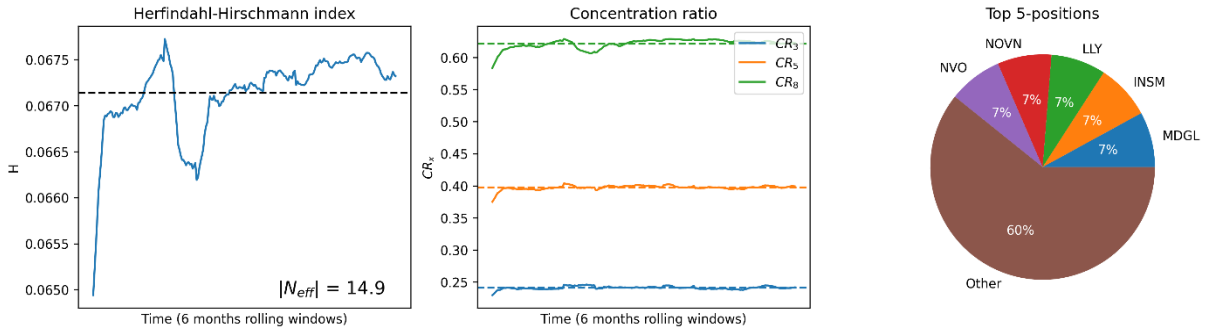


Figure 6 - Concentration metrics. Herfindahl-Hirschman Index and concentration ratios (CR3, CR5, CR8) calculated over rolling windows. Results indicate a diversified portfolio, with no persistent dominance of individual positions due to position-level constraints.

Concentration metrics confirm that the portfolio remains well diversified throughout the period. Position-level limits effectively prevent excessive exposure to individual holdings, while industry constraints reduce the risk of sector concentration. Quantitative indicators, including the Herfindahl-Hirschman Index and concentration ratios, remain consistent with a diversified allocation, which is particularly important in a thematic strategy exposed to high-growth segments.

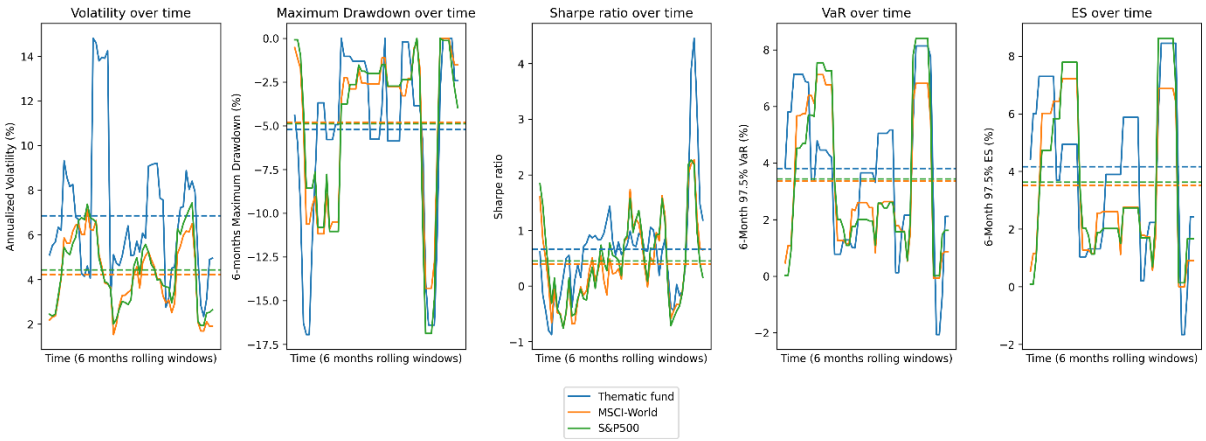


Figure 7 - Risk metrics over time. Rolling annualized volatility, maximum drawdown, Sharpe ratio and tail-risk measures. The portfolio exhibits higher volatility than broad benchmarks, consistent with its thematic positioning, while downside risk remains broadly aligned with the defined constraints.

Risk analysis shows that the portfolio exhibits higher volatility than broad equity indices, reflecting its exposure to innovation-driven companies and event-driven dynamics such as clinical outcomes and regulatory developments. However, downside risk remains broadly consistent with the constraints imposed. The one-month 97.5% Value at Risk remains within the targeted range for most of the analyzed period, indicating that tail-risk controls are effective, although not always binding.

α	0
β_{market}	0.6
β_{theme}	0.7
Tracking errror (%)	0.3
R^2	1

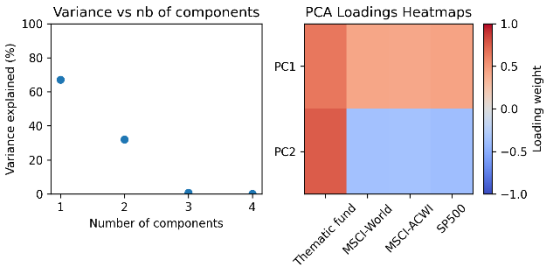


Figure 8 - Principal Component Analysis (PCA). Decomposition of portfolio and benchmark returns into orthogonal factors. The first component represents broad market exposure, while the second captures theme-specific dynamics, supporting the presence of a differentiated cardiometabolic innovation factor.

Principal Component Analysis provides additional insight into the drivers of performance. The first component is associated with general market exposure, while the second captures a more theme-specific factor. The portfolio's exposure to this second component suggests that returns were not solely driven by broad equity market movements, but also by the cardiometabolic innovation theme.

Table 1 - Performance comparison with benchmarks. Comparison of return and risk metrics between the thematic portfolio and global indices (MSCI World, MSCI ACWI, S&P 500). The portfolio demonstrates higher returns with increased but controlled volatility, consistent with a thematic equity strategy.

	THEMATIC FUND	MSCI-WORLD	MSCI-ACWI	S&P-500
ANNUALIZED RETURNS (%)	24.3	4.9	4.2	6.5
ANNUALIZED VOLATILITY (%)	19.3	11.8	11.4	12.4
MAXIMUM DRAWDOWN (%)	-16.9	-19.5	-20.3	-16.9
SHARPE RATIO	1.3	0.4	0.4	0.5
1-MONTH 97.5% VAR (%)	6.6	6.6	6.5	7.3
1-MONTH 97.5% ES (%)	7.9	7.0	6.8	8.2

Overall, the results indicate that the portfolio was able to generate excess returns relative to global benchmarks while maintaining diversification and risk controls consistent with a structured thematic investment approach. These findings should, however, be interpreted with caution, as the analyzed period benefited from favorable sector-specific dynamics which may not persist in the future.

Performance Drivers

Portfolio performance is primarily driven by innovation cycles in cardiometabolic therapies, particularly in areas with significant unmet medical need. The commercial adoption of new treatments, regulatory developments and reimbursement decisions are key drivers of performance. In particular, the expansion of reimbursement coverage and broader adoption of new therapies can significantly influence market dynamics. Risks include uncertainty around long-term clinical outcomes, regulatory changes, pricing pressure and the potential for scientific setbacks. The relatively short history of certain therapeutic classes also introduces uncertainty regarding long-term effectiveness and safety.

The portfolio's diversified exposure across the cardiometabolic ecosystem is intended to mitigate reliance on any single driver.

ESG & Impact Considerations

Cardiometabolic innovation has clear societal relevance, particularly in addressing diseases such as obesity and diabetes. However, it also raises ethical and reputational considerations. For example, the use of obesity treatments for non-medical purposes has generated public debate. Pricing, access and advertising practices may also create regulatory and reputational risks. These considerations highlight the importance of integrating ESG awareness into thematic investing, particularly in sectors with significant societal impact.

Conclusion

This project demonstrates how a scientific background can be translated into an investment framework. Cardiometabolic innovation was selected because it combines strong long-term medical demand, active scientific innovation and a broad investable ecosystem.

The analysis follows a full investment process: theme definition, investable-universe construction, company-level thematic scoring, equity selection, constrained portfolio optimization, transaction-cost assessment, risk analysis and benchmark comparison. The resulting portfolio showed strong historical performance relative to global equity benchmarks, with higher but controlled volatility and limited turnover.

The results should be interpreted with appropriate caution. The period analyzed benefited from strong thematic tailwinds, particularly around GLP-1-related equities. Future returns may be lower, and the sector remains exposed to clinical, regulatory, reimbursement and valuation risks. These limitations reinforce the importance of diversification, transparent assumptions and robust risk controls.

For me, the project was valuable because it required the integration of domain expertise, quantitative modelling and investment judgement. It strengthened my conviction that my background in biophysics can be relevant in finance, particularly in thematic equity research, healthcare investing, portfolio analytics and data-driven investment processes.